

KH26-ECE-11 – AI-Based Smart Microgrid Energy Management System

Background

Campuses and industries increasingly use a combination of grid electricity, solar energy, battery storage and backup generators. Poor coordination among these sources can increase electricity costs and cause unnecessary energy wastage.

Problem Statement

Develop an intelligent microgrid-management platform that monitors power generation, consumption, battery level and grid availability. The system should predict short-term energy demand and recommend the most efficient combination of available power sources. It should prioritize renewable energy, prevent battery overcharging and generate alerts during overload or power failure. Administrators should be able to monitor energy flow, cost and carbon-emission reduction through a dashboard.

Mandatory Requirements

- Multiple energy-source monitoring
- AI-based load prediction
- Renewable-energy prioritization
- Battery and grid status tracking
- Overload and failure alerts
- Energy-cost and carbon analytics

Expected Prototype

A working dashboard demonstrating intelligent energy-source selection using simulated or sensor-generated power data.

KH26-ECE-12 – Intelligent EV Charging and Grid Load Coordination System

Background

Simultaneous charging of many electric vehicles can create peak demand, long waiting times and excessive stress on the electrical grid. Charging stations require a coordinated mechanism for scheduling vehicles efficiently.

Problem Statement

Develop an AI-enabled EV charging platform that assigns charging slots based on battery level, vehicle priority, charger availability, electricity demand and user departure time. The

system should prevent grid overload and dynamically reschedule vehicles when demand changes. Users should be able to reserve charging slots and monitor charging progress. Administrators should receive station-utilization statistics and peak-load alerts.

Mandatory Requirements

- EV registration and slot booking
- AI-based charging prioritization
- Grid-load balancing
- Dynamic schedule adjustment
- Charging status tracking
- Station-utilization dashboard

Expected Prototype

A functional application demonstrating EV slot allocation and grid-load management using simulated charging-station data.

KH26-ECE-13 – AI-Enabled Transformer Health Monitoring and Fault Prediction

Background

Distribution transformers may fail because of overheating, overloading, insulation deterioration or abnormal electrical conditions. Unexpected transformer failure can interrupt power supply and increase maintenance costs.

Problem Statement

Develop an intelligent transformer-monitoring system that analyses voltage, current, temperature, oil level and load information. The system should identify abnormal conditions, estimate transformer health and predict possible faults before breakdown. Maintenance staff should receive early warnings and recommended inspection priorities. A dashboard should display transformer status, historical trends and locations requiring urgent attention.

Mandatory Requirements

- Electrical and environmental parameter monitoring
- AI-based fault-risk prediction
- Transformer health score
- Critical-condition alerts
- Maintenance-priority recommendation
- Historical trend visualization

Expected Prototype

A dashboard or IoT prototype demonstrating transformer-health analysis using simulated or sensor-generated data.

KH26-ECE-14 – Edge-AI Railway Track and Obstacle Monitoring System

Background

Cracks, fallen objects, animals and other obstacles on railway tracks can create serious safety risks. Continuous manual inspection is difficult, particularly across remote track sections.

Problem Statement

Develop an edge-AI monitoring system that analyses camera or sensor input to identify possible track defects and obstacles. The solution should classify the detected risk, estimate its location and generate an immediate alert. It should continue basic operation even when internet connectivity is limited and synchronize records when the connection is restored. Railway personnel should be able to review alerts and inspection history through a dashboard.

Mandatory Requirements

- Track-image or sensor-data processing
- AI-based defect and obstacle detection
- Risk classification and alert generation
- Location or section identification
- Offline or edge-processing capability
- Inspection-history dashboard

Expected Prototype

A working proof of concept using sample track images, prerecorded video or simulated sensor data.

KH26-ECE-15 – AI-Based Adaptive Traffic Signal Coordination Network

Background

Traffic signals operating with fixed timings cannot respond effectively to varying traffic density, emergency vehicles or temporary road congestion. Independent control of nearby signals may also shift congestion from one junction to another.

Problem Statement

Develop an intelligent traffic-signal coordination system that estimates vehicle density from cameras or sensors and dynamically adjusts signal timings. The system should coordinate multiple junctions, give controlled priority to emergency vehicles and avoid excessive waiting for low-traffic roads. Administrators should be able to monitor congestion levels and manually override signal recommendations when necessary.

Mandatory Requirements

- Vehicle-density estimation
- Dynamic signal-time calculation
- Multi-junction coordination
- Emergency-vehicle prioritization
- Manual override facility
- Traffic analytics dashboard

Expected Prototype

A simulation or hardware model demonstrating adaptive signal control across at least two connected junctions.

KH26-ECE-16 – Distributed Flood and Landslide Early-Warning Network

Background

Floods and landslides may occur rapidly in vulnerable areas, while conventional monitoring systems often depend on centralized infrastructure. Communication failure during extreme weather can delay warnings to nearby communities.

Problem Statement

Develop a distributed warning network using water-level, rainfall, soil-moisture or vibration data. The system should combine sensor readings with AI-based risk assessment to classify the situation as normal, warning or critical. Nearby nodes should continue sharing alerts even when the central internet connection fails. Authorities should receive location-wise risk information, while residents should receive simple warning messages.

Mandatory Requirements

- Multi-sensor data collection
- AI-based risk classification
- Local node-to-node communication
- Offline alert capability
- Location-wise warning dashboard

- Public alert generation

Expected Prototype

A sensor-based or simulated network demonstrating risk detection and alert communication under normal and disconnected conditions.

KH26-ECE-17 – AI-Based Industrial Worker Safety and Hazard Response System

Background

Industrial environments may expose workers to hazardous gases, excessive temperature, restricted zones and unsafe machine conditions. Delayed detection can lead to injuries and equipment damage.

Problem Statement

Develop an intelligent safety system that combines environmental sensors, wearable information or camera input to identify hazardous conditions. The platform should assess risk severity, alert nearby workers and recommend actions such as evacuation, ventilation or machine shutdown. Supervisors should be able to view worker location, hazard history and unresolved incidents. The system must respect privacy and use only authorized monitoring.

Mandatory Requirements

- Environmental or wearable-data monitoring
- AI-based hazard classification
- Worker and supervisor alerts
- Safety-action recommendations
- Incident acknowledgement and tracking
- Hazard analytics dashboard

Expected Prototype

A working safety-monitoring prototype using simulated sensor readings or low-cost hardware components.

KH26-ECE-18 – AI-Assisted Precision Agriculture and Automated Irrigation System

Background

Farmers may apply water and fertilizers uniformly even though soil moisture, crop condition and weather vary across different areas of a field. This results in resource wastage and reduced productivity.

Problem Statement

Develop an intelligent agriculture system that analyses soil moisture, temperature, humidity, weather and crop-condition information. The system should recommend or automatically control irrigation for different field zones. It should detect abnormal crop conditions and provide simple alerts or advisory messages. Farmers should be able to monitor field status and manually control pumps whenever required.

Mandatory Requirements

- Multi-parameter field monitoring
- AI-based irrigation recommendation
- Zone-wise water control
- Crop-condition alerts
- Manual pump override
- Farmer-friendly dashboard

Expected Prototype

A hardware or simulation-based prototype demonstrating automated irrigation and field-condition monitoring.

KH26-ECE-19 – AI-Based Non-Intrusive Appliance Energy Analytics System

Background

Consumers usually receive only total electricity-consumption information and cannot identify which appliances are responsible for high usage. Installing separate meters for every appliance can be costly.

Problem Statement

Develop a non-intrusive energy-monitoring solution that estimates appliance-level electricity usage from combined power-consumption patterns. The system should identify major energy-consuming appliances, detect unusual usage and recommend energy-saving actions. Users should be able to compare daily or weekly consumption and receive alerts when expected energy limits are exceeded.

Mandatory Requirements

- Combined power-data analysis

- AI-based appliance-usage estimation
- Abnormal-consumption detection
- Energy-limit alerts
- Usage comparison reports
- Personalized energy-saving recommendations

Expected Prototype

A software or sensor-assisted prototype demonstrating appliance-level estimation using sample or collected power data.

KH26-ECE-20 – Resilient Emergency Communication System for Disaster Zones

Background

During cyclones, floods and earthquakes, cellular and internet networks may become unavailable. Rescue teams and affected communities still require a basic method to exchange location, distress and resource information.

Problem Statement

Develop a resilient communication prototype that allows nearby devices or low-power nodes to exchange short emergency messages without depending entirely on internet connectivity. The system should prioritize distress messages, maintain a store-and-forward queue and identify the last known location of the sender. An AI component should classify incoming messages by urgency and group duplicate requests. A control-room interface should display received messages and communication status.

Mandatory Requirements

- Device-to-device or node-to-node communication
- Offline message storage and forwarding
- AI-based urgency classification
- Duplicate-request grouping
- Location and sender tracking
- Control-room monitoring dashboard

Expected Prototype

A simulated or hardware-assisted communication network demonstrating emergency-message exchange during internet failure.